

Meaghan Ramlakhan

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EDUCATION

Rice University

Houston, TX

B.S. in Computer Science; Minors in Data Science and Statistics; GPA: 3.8/4.0

Expected May 2026

- Relevant Coursework: Intro to Program Design, Reasoning About Algorithms, Computer Engineering, Linear Algebra, Computer Systems, R for Data Science, Concurrent Program Design
- Leadership & Organizations: Orientation Week Co-Advisor, GHC 2025, Rice Women in STEM Mentor, HackRice 14 PR Board, CSters, Rice Caribbean Students Society, Women Impact Tech 2024, DEEP Student Mentor

EXPERIENCE

CPRIT BIG-TCR Cancer and Artificial Intelligence Summer Intern

June 2025 – August 2025

McWilliams School of Biomedical Informatics at UTHealth Houston

Houston, TX

- Engineered and optimized a 5-layer MLP neural network in Python/TensorFlow for antibody-protein binding prediction, applying cross-validation, L2 regularization, and hyperparameter tuning to achieve 85%+ accuracy.
- Extended **BRAINCELL-AID**, a biomedical brain annotation web platform, by designing Python/MySQL pipelines for automated data ingestion, processing 5000+ cell records, and building interactive data visualizations to support 100+ users.

Computer Science Teaching Assistant, Rice University

August 2023 – Present

Current - Concurrent Program Design (COMP 318)

Houston, TX

- Served as a Teaching Assistant for four Computer Science courses, supporting 800+ students through office hours, Piazza responses, and assessment grading.
- Managed five groups of three students in COMP 318, guiding them through the design and implementation of two major full-stack software projects over the semester.
- Met regularly with project teams to provide technical direction, ensure progress milestones were met, and foster collaboration in concurrent program design.

Live Instructor Technology Intern - RCEL ELITE Tech Program

May 2024 – July 2024

Rice University

Houston, TX

- Selected as one of 20 instructors, to design and refine canvas and course materials on machine learning and robotics, collaborating closely with Head Instructors to ensure high-quality content delivery.
- Led 40+ classroom sessions on cutting-edge machine learning and robotics topics, delivering interactive lessons, detailed feedback, and managing attendance and grading with accuracy.

Computer Hardware Engineer & Mission Assurance Specialist

May 2023 – August 2023

NASA L'SPACE Program - ASU

Remote (Headquartered in Arizona)

- Collaborated with 11 undergraduate students nationwide to conceptualize a cost-effective robotic mission, enhancing understanding of Ceres' geological attributes and habitability potential.
- Designed intricate Command and Data Handling subsystems, developed comprehensive software architecture and risk management strategies, and delivered detailed documentation and risk analysis triweekly, aligning with NASA's meticulous standards.

PROJECTS

OwIDB: A NoSQL Document Database | Golang, RESTful API, HTTP Protocols

Aug–Oct 2024

- Built a concurrent NoSQL DB with CRUD REST APIs, skip-list indexing, schema validation, and live client updates via SSE.

Comparative Analysis of LSH Methods | Python, Scikit-learn

Sep–Dec 2024

- Benchmarked Locality Sensitive Hashing variants on large corpora to assess speed and accuracy tradeoffs in text similarity.

RDT Ticketing App | MongoDB, HTML/CSS, JS, Node.js

Jan–Apr 2024

- Designed and deployed a full-stack platform for Rice Dance Theatre to manage ticket sales, show listings, and seat availability.

FDA Adverse Events Analysis | Python, Pandas, Scikit-learn

Jan 2024

- Won Best Underclassmen Award for modeling risks in FDA's CAERS dataset to flag hazardous consumer products.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS, Golang, Typescript, C, R

Frameworks: React, Node.js

Developer Tools: Git, VS Code, Visual Studio, PyCharm, IntelliJ, AWS (Lambda, API Gateway, S3, RDS, DynamoDB), Linux (Command Line), Microsoft Office (Excel, Word, Powerpoint)

Libraries: Pandas, NumPy, Matplotlib, Sci-Kit Learn, Pytorch, Tensorflow

Certifications: Break Through Tech Certificate in Machine Learning Foundations (Cornell University August 2025), Paycom Technology Summer Engagement Program (July 2023), Introduction to Programming using Java (Great Learning June 2023), The Data Science Course: Complete Data Science Bootcamp (Udemy May 2023)